Project Proposal

Mili Shah

```
library(tidyverse)
library(tidymodels)
library(readr)
library(vroom)
library(patchwork)
library(knitr)
```

```
# load dataset

# vroom function from:
# https://cran.r-project.org/web/packages/vroom/vroom.pdf#:~:text
# =include_na%20If%20TRUE%20and%20the%20data%20contains,A%20format%20specification.
# %20If%20set%20to%20%22%22:

food_access <- vroom("data/Food Access Research Atlas.csv",
                     na = c("", "NA", "NULL"),
                     show_col_types = FALSE)

# load data dictionary to paste into data README.md
data_dict <- read_csv("data/VariableLookup.csv", show_col_types = FALSE)
```

Introduction

Section 1: Introduction

- An introduction to the subject matter you're investigating (citing any relevant literature)
- Statement of a well-developed research question.
- The motivation for your research question and why it is important
- Your hypotheses regarding the research question

– This is a narrative about what you think regarding the research question, not formal statistical hypotheses.

Food

Data description

The 2019 Food Access Research Atlas that I will be analyzing is from the USDA Economic Research Service. It provides an overview of food access indicators in the US by measuring variables like supermarket accessibility, race, poverty rate, and median income according to census tracts (Food Access Research Atlas - About the Atlas). This data set is a combination of data collected by the 2010 Decennial Census, the 2014-18 American Community Survey (ACS), and two 2019 lists of supermarkets, one being from Trade Dimensions TDLinx. The 2010 Census provided population data relating to age, race, income, poverty rate, and residence in group quarters. The 2014-18 ACS contributed information about income, vehicle availability, and SNAP participation. Finally, two lists of grocery stores were combined to produce a list of stores that have nutritious and affordable food. This data was then used to compute the variables that communicate the distance that one lives from a grocery store ("Food Access Research Atlas - Documentation").

Each observation in the data represents an individual census tract measured by the 2010 US Census and uniquely identified by the `CensusTract` variable. Generally, this dataset measures food store access for individuals and for neighborhoods, using different variables to measure low income and low access communities that lack access to healthy food sources. Most of the measures include measures of accessibility to sources of healthy food (by distance to a store), individual resources that impact accessibility (vehicle availability or income), and neighborhood level indicators of resources (public transporation, average income) ("Food Access Research Atlas - About the Atlas").

Data processing

This data set currently has 72,531 observations representing the number of Census tracts in the 2010 U.S. Census. Training a regression model on too many observations, however, poses a risk of overfitting. The model has the potential to learn random noise and thereby ignore the true pattern underlying the data. It can also make statistical significance tests too sensitive, resulting in "statistically significant" differences when there were likely not any. To address this issue, I will take a random sample of 500 observations from the data set. This smaller sample via randomization also seeks to avoid the issue of spatial dependence, since census tracts near each other are likely to have similar data.

In addition to reducing the size of my data set, I also plan to convert the indicator predictor variables of interest to categorical variables. Currently, the data has marked each variable

except State and County as numeric. However, variables like `Urban`, which simply display if a tract is urban or not, should be represented as a factor with two levels. Once I convert these variables into categorical predictors, my regression model and data visualizations will be interpretable.

Finally, I will address the issue of missing data. There are 29,957 missing observations in the `LALOWI1_10` predictor variable alone. In order to address the number of missing observations, I plan to drop all the n/a values in my data set. I propose taking this approach rather than setting the n/a values to 0 because in cases like with `LALOWI1_10`, a value of 0 represents that 0 people in a Census tract were considered low income and living far from a supermarket. This is not entirely plausible, as it may be the case that a tract does not have data simply because they were not sampled. Furthermore, since so many values are missing from the data set, converting them all to 0 could significantly skew the model's predictions.

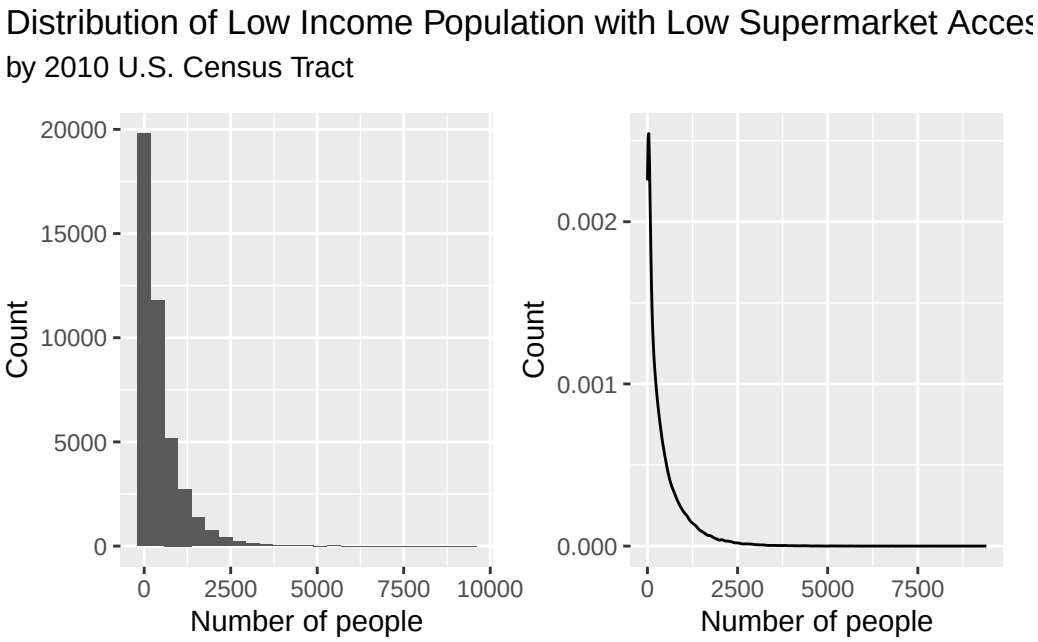
Analysis approach

```
# summary statistics
summary <- food_access |>
  summarise(
    min = min(LALOWI1_10, na.rm = TRUE),
    Q1 = quantile(LALOWI1_10, 0.25, na.rm=TRUE),
    median = median(LALOWI1_10, na.rm = TRUE),
    mean = mean(LALOWI1_10, na.rm = TRUE),
    Q3 = quantile(LALOWI1_10, 0.75, na.rm=TRUE),
    sd = sd(LALOWI1_10, na.rm = TRUE),
    IQR = Q3 - Q1,
    max = max(LALOWI1_10, na.rm = TRUE),
    # following line from https://stackoverflow.com/questions/29076536/
    # how-can-i-count-number-of-na-values-in-dataset
    n_missing = sum(is.na(LALOWI1_10))
  )

# visualize distribution of response var LALOWI1_10 - ind food access
p1 <- food_access |>
  ggplot(aes(x=LALOWI1_10)) +
  geom_histogram(bins=25) +
  labs(x = "Number of people", y = "Count")

p2 <- food_access |>
  ggplot(aes(x=LALOWI1_10)) +
  geom_density() +
```

```
  labs(x = "Number of people", y = "Count")

summary
```

```
# A tibble: 1 x 9
    min    Q1 median  mean    Q3    sd   IQR   max n_missing
  <dbl> <dbl>  <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>     <int>
1     0    49    228  442.   605  589.   556  9402     29957
```

```
plots <- p1 + p2
plots + plot_annotation(
  title = "Distribution of Low Income Population with Low Supermarket Access",
  subtitle = "by 2010 U.S. Census Tract"
)
```



The response variable `LALOWI1_10` represents the "low-income population count beyond 1 mile for urban areas or 10 miles for rural areas from [the] supermarket" (data dictionary). The USDA defines a food desert as a low-income census tract where a substantial amount of residents lack adequate access to healthy and affordable food. Specifically, census tracts meet this criteria when low income is defined as a poverty rate of 20% or higher and when low access is defined by at least 500 people or at least 33% of the population living more than 1 mile from a supermarket in urban areas or more than 10 miles in rural areas. This definition matches

that of `LALOWI1_10` , so I plan on using `LALOWI1_10` to represent the number of people in a census tract living in a food desert.

The histogram displays a unimodal, right-skewed distribution, indicating that most census tracts have relatively small low-income populations living far from a supermarket. The long right tail suggests that while a small number of tracts have very high counts, the majority are concentrated at lower values. Due to the skew, the mean is not an accurate indicator of the center of the distribution. Thus, the median of 228 suggests that in a typical census tract, about 228 low-income residents live beyond 1 mile (urban) or 10 miles (rural) from a supermarket. The spread of the data is about 556 and most of the data (middle 50%) is between 49 people and 605 people living in a food desert in a census tract. There is a minimum of 0 people and a maximum of 9402 people in a census tract living in a desert track. There are a couple outliers in the data at census tracts with more than 4000 low-income people living with little access to healthy food. Though the histogram doesn't show these outliers, we can look to the density plot and the summary statistics to understand that outliers exist outside the bins on the histogram plot.

To analyze the factors that influence the size of the low-income population living far from healthy food (`LALOWI1_10`), I plan to examine `PovertyRate` (poverty rate in a census tract), `Urban` (if the tract is in an urban or rural area), `TractSNAP` (number of housing units receiving SNAP benefits in a tract), and`TractHUNV` (total number of housing units without a vehicle). Since my response variable is quantitative, I will fit a multiple linear regression model to understand how the predictors are associated with a difference in the number of low-income people living far from a supermarket in a census tract. These predictors were selected because literature shows that food deserts are characterized by factors like higher poverty, small populations, lower educational attainment, and less transportation access, to name a few ("Characteristics and Influential Factors of Food Deserts"). Informed by these findings, I have selected predictors like `PovertyRate` and `TractHUNV` . I chose to add `TractSNAP` because I am interested in seeing if receiving SNAP benefits is a good proxy for food insecurity. Additionally, many studies analyzed how the minority population influenced the population in a food desert. I chose `Urban` to see if the population density played a role instead. While the data contains many variables that mean the same thing, just measured the same way – for instance, a population amount or proportion – I chose all of the quantitative predictor variables to represent a count of individuals instead of a proportion to be consistent with the way the response variable is measured.

Data dictionary

The data dictionary can be found here [Update the link and remove this note!]

AI Disclosure

- I asked AI about an error or why code would do something I didn't want.